

Daniel Gregory

New Lenox, IL | Phone: (708) 205-0688

Email: dannytgregory10@gmail.com | Portfolio: <https://danielgregory.github.io>

Summary

Underwriting analytics analyst at PURE Insurance, a high net worth personal lines P&C insurer, turning complex underwriting and policy data into recommendations for executive decisions on competitive positioning, retention, and underwriting requirements. Combines an MS in Computer Science and a BS in Mathematics with hands-on SQL, Python, and modeling experience.

Professional Experience

Analyst, Underwriting Analytics, PURE Insurance – *Chicago, IL* (Jan 2026–Present)

- Delivered competitor benchmarking to the C-suite, including the CEO, showing PURE's performance relative to peers and identifying where to focus attention.
- Presented a deep dive on member and policy retention to the Chief Financial Officer and Chief Growth Officer, confirming the book was strengthening as weaker business ran off amid a softer market.
- Analyzed protection credits and showed central station alarm requirements could be adjusted in line with competitor changes, informing changes now in implementation and review.
- Built a broker reassignment simulator in SQL and Python, adopted by underwriting leadership, significantly reducing the time to reassign brokers and plan after underwriting changes.
- Transformed raw backend Dragon data into SQL quote version history tables, creating the data foundation for underwriting risk analysis and becoming the team's lead expert on Dragon data.

Numerical Analysis Research Assistant, University of Kentucky – *Lexington, KY* (Jun 2023–Aug 2024)

- Developed a MATLAB-based solver for convection-diffusion problems using Shishkin meshes, roughly halving max-norm error under mesh refinement.

Data Analytics Trainee, The Global Tech Experience – *Remote* (May 2023–Aug 2023)

- Interpreted data trends via Tableau visualizations, supporting strategic decisions across teams.
- Built an A/B testing framework to optimize email engagement, improving response metrics.

Education

University of Illinois Urbana-Champaign, Champaign, IL (2024–2025)

- **Master of Computer Science**
- GPA: 3.70
- Relevant Courses: Advanced Bayesian Modeling, Applied Machine Learning, Data Science

University of Kentucky, Lexington, KY (2020–2024)

- **Bachelor of Science in Mathematics** — Minors: Computer Science, Psychology
- GPA: 3.50, Cum Laude | Bluegrass Spirit and UK Futures Scholarship Recipient
- Relevant Courses: Probability, Statistical Methods, Numerical Methods

Relevant Projects

AI-Driven Trading System – *Personal Project* (Dec 2024–Present)

- **Tech:** Python (pandas, TensorFlow, Scikit-learn), GRU sequence modeling, reinforcement learning (PPO), automated retraining.
- Built a weekly pipeline that retrains on up to 100 stocks, combining a GRU model predicting next-week direction, an SGD classifier validating that signal, and a PPO agent rewarded for risk-adjusted returns over 200 days of backtest.
- Ranks bullish candidates by a composite relative-strength score (cross-sectional percentiles of 20-, 60-, and 90-day returns) and allocates capital across the top 5, capped at 25% per position.
- Monitors held positions daily, automatically exiting any that flip bearish and replacing them with the next-ranked candidate.

Skills

- **Languages:** Python, SQL, R, MATLAB, Java, C++
- **Python libraries:** NumPy, pandas, XGBoost, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn
- **Data & AI:** Machine/Deep Learning, Predictive Modeling, Time-series Analysis, A/B Testing
- **Tools:** Databricks, Hex, Microsoft Excel, Tableau, AWS, Git
- **Domain:** High Net Worth P&C Insurance, Underwriting Analytics, Retention and Competitive Analysis